Empathetic Cognitive Support Agent (ECSA)

- AI-Powered Intelligent Customer Support Chatbot
  - Artificial Intelligence Capstone Project
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Problem Statement

- Traditional chatbots lack emotional intelligence.
  - Rigid systems fail to adapt tone dynamically.
  - Need for context-aware and human-like AI support.

Project Objective

- Develop transformer-based intent detection.
  - Implement emotion-aware adaptive responses.
  - Integrate confidence-based escalation logic.

System Architecture

- User Input Processing
  - Intent Detection (Transformer Model)
  - Emotion Analysis (Sentiment Layer)
  - Adaptive Response Engine
  - Confidence Evaluation & Escalation

Technology Stack

- Python
  - Google Colab
  - Sentence-Transformers
  - TextBlob
  - NumPy & Pandas

Intent Detection Mechanism

- User query converted to embeddings.
  - Cosine similarity for semantic matching.
  - Highest similarity determines intent class.

Emotion Detection Layer

- Sentiment polarity analysis.
  - Classified as Positive / Neutral / Negative.
  - Tone adapts based on detected emotion.

Adaptive Response Engine

- Emotion-based tone selection.
  - Context-aware reply generation.
  - Human-like conversational behavior.

Confidence-Based Escalation

- Similarity score calculated.
  - If confidence < threshold → escalate to human.
  - Prevents incorrect automated guidance.

Humanization Features

- Typing simulation effect.
  - Emotionally empathetic wording.
  - Conversational continuity handling.

Sample Conversation Output

- Live chatbot interaction with emotion detection.
  - Confidence score displayed with each reply.

Performance Highlights

- Semantic understanding using transformers.
  - Emotion-aware intelligent adaptation.
  - Dynamic decision-making capability.

Project Implementation Link

- Google Colab Notebook:
  - https://colab.research.google.com/drive/1KDilDkwq7lM0Jxg9qbi9u-k9BO-MwOIe?usp=sharing

Applications

- E-commerce platforms
  - Banking customer support
  - Technical support systems
  - Enterprise helpdesk automation

Conclusion & Future Scope

- Enhances user satisfaction using Emotional AI.
  - Can be deployed as web or API service.
  - Future: Add multilingual & voice interaction.